

Modelling the Fate and Trajectories of Hypothetical Oil Spills Near Chattogram and Mongla Ports, Bangladesh

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Abstract

Oil spills pose substantial threats to marine ecosystems, particularly in coastal zones with intense shipping activities. The northern Bay of Bengal, especially the area around Chattogram and Mongla ports, faces a growing risk from increased vessel and oil shipments. This study, Bangladesh's first scenario-based oil spill model, uses the OpenOil module of the OpenDrift framework to simulate hypothetical oil spill incidents under varying seasonal conditions. Two locations offshore of Chattogram and Mongla ports were selected for simulation during the summer and winter monsoon of 2024. High-resolution ocean and atmospheric data from the Copernicus Marine Environment Monitoring Service (CMEMS) to simulate the fate of 10,000 oil particles over a span of 7 days. Wind, current, wave along with weathering mechanisms of oil, i.e., evaporation, emulsification, and biodegradation of a Medium Crude Oil (API 30), were taken into account. The results showed significant seasonal variations in oil spill fate and coastal stranding. During the summer monsoon, strong winds (6-12 m/s) and currents at Chattogram drove approximately 48% of oils to the shoreline, while at Mongla, all the droplets got beached within five days, indicating huge coastal exposure risk. At winter monsoon, nearly all oil particles at Chattogram (99%) were stranded, however, no stranding was observed at Mongla, with oil slick moved in the direction of south-southwest due to moderate wind (4-7 m/s) and currents (0.1-0.2 m/s) and thereby 60% of oils afloat after a week. Weathering processes varied seasonally. Summer led to active weathering (>70% oil biodegraded near Chattogram, >95% emulsification at Mongla). Winter had weak weathering (<30% oil biodegraded, 70% emulsification), increasing persistence and offshore movement. These findings highlight the need for season-specific response strategies to protect Bangladesh's sensitive coastal zones. Simulation accuracy can be improved with higher-resolution datasets.

Keywords: *Oil Spill Modelling, OpenOil, Fate of Oil Spills, Chattogram and Mongla Ports*